

Epithelial-Mesenchymal Transition Spectrum Quantification and its Efficacy in Deciphering Survival and Drug Responses of Cancer Patients

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Supplementary Text

Verifying bladder cancer-specific EMT signature

Breast and ovarian cancer-specific EMT signatures were previously verified (Akalay *et al*, 2013; Miow *et al*, 2014) and assessed again in Fig 1C. Hence, because of the relative lack of data in other cancers studying EMT parameters of multiple cell lines, we focused on verifying the bladder cancer-specific EMT signature (Fig S1). The EMT scores of bladder cancer cell lines were computed based on microarray gene expression of BLA-40 (Lee *et al*, 2007) and CCLE (Barretina *et al*, 2012). Data for the immunofluorescence staining of EMT markers, cell line morphology and invasion assays were taken or inferred from (Baumgart *et al*, 2007; Black *et al*, 2008; Chen *et al*, 2009) based solely on cell line name. Consistent with breast cancer data (Fig 1C), cell lines with high positive staining for VIM and CDH2 (Thiery *et al*, 2009) had a significantly greater EMT score ($p=0.0324$ and $p=0.0022$, respectively; Fig S1). Conversely, cell lines with high positive staining for known Epi markers, CDH1 and Plakoglobin, had a significantly lower EMT score ($p=0.0025$ and $p=0.0062$, respectively). No nuclear localisation of β -catenin was found in cell lines with high CDH1 staining intensity (Black *et al*, 2008). Importantly, not only did the EMT score provide a good indication as to whether a cell line had an Epi or Mes morphology ($p=7.54E-4$), but it could identify any intermediate phenotypic cell lines that displayed both Epi and Mes morphologies. The invasion assay also affirmed that cell lines with higher EMT scores were more invasive ($p=0.0016$). However, one caveat of this approach was that these experimental data were taken or inferred and combined from publications based solely on cell line names: it is possible that different cell lines were labelled with the same name, or the same cell line was cultured differently by different laboratories. In addition, we found one discrepancy for the highly metastatic UMUC14 bladder cancer cell line (Karashima *et al*, 2003): Chen and colleagues previously reported that it has an Epi morphology (Chen *et al*, 2009); yet, our EMT scoring revealed otherwise. Nevertheless, these results indicate the overall congruency between the bladder cancer-specific EMT signature scores and the published literature.

Verifying Generic EMT signature on Pancreatic Cancer

EMT has been implicated in pancreatic cancer progression and drug resistance (Arumugam *et al*, 2009; Hotz *et al*, 2007; Nakajima *et al*, 2004). Since the generic EMT signature was not derived from a pancreatic cancer-specific EMT signature, it was important to ascertain that the generic EMT scoring was capable of quantitating the EMT status accurately in pancreatic cancer. In Fig S3, we have validated the applicability of the generic EMT signature in pancreatic cancer where the change in EMT status of a pancreatic cancer cell line, PANC-1, induced with TGF β , was accurately reflected. To further study the applicability of the generic EMT signature in pancreatic cancer, we performed additional validation of the generic EMT scoring in this cancer type. Fig S4 shows the relative protein expression normalized to β -actin, as inferred from western blot (Hotz *et al*, 2007; Zhu *et al*, 2012) (Supplementary Materials and Methods), immunofluorescence (IF) staining, and migration assays (Arumugam *et al*, 2009) of seven pancreatic cancer cell lines. EMT scores were computed based on gene expression data from SANGER COSMIC (Garnett *et al*, 2012) and CCLE (Barretina *et al*, 2012) collections. Note that the experiment data from publications was matched to the computed EMT score based solely on the cell line name. We observed that well-differentiated cell lines CAPAN-1 and HPAF-II were identified as Epi whereas undifferentiated cell lines MIAPaCa-2 and PANC-1 were identified as Mes. Western blot analysis of Epi cell lines showed strong relative protein expression of CDH1, the prototypic adhesion molecule of epithelial cells, whereas the relative expression of Mes markers (CDH2, SNAI1, SNAI2, TWIST1) were not limited to Mes cell lines (Hotz *et al*, 2007). However, positive IF for ZEB1 and VIM (Arumugam *et al*, 2009) was observed for the Mes cell lines, affirming the accuracy of generic EMT scoring. Migration assays conducted previously (Arumugam *et al*, 2009) uphold the belief that Mes cell lines are more invasive, a finding unambiguously reflected by the EMT score.

Generic EMT signature and miRNA

Expression levels of miRNAs have been implicated in promoting and suppressing EMT (Hao *et al*, 2014; Lim & Thiery, 2012; Zhang & Ma, 2012). In order to investigate whether a generic EMT

signature for miRNA can be established, we collected six datasets from GEO and TCGA comprising mRNA and miRNA expression profiles. We correlated the EMT score computed from the mRNA generic EMT signature with the expression of EMT-regulating miRNA (Hao *et al*, 2014; Zhang & Ma, 2012) in bladder (GSE40355) (Hecker *et al*, 2013), pancreas (GSE32688) (Donahue *et al*, 2012), prostate (GSE21034) (Taylor *et al*, 2010), breast (TCGA) (The Cancer Genome Atlas, 2012), ovarian cancer (TCGA) (The Cancer Genome Atlas, 2011) and multiple myeloma (GSE17498; Fig S2) (Lionetti *et al*, 2009). While it appears that miRNAs regulating EMT are cancer-specific, miR-200 (miR-200a, miR-200b, miR-200c, miR-141, miR-429) and miR-34 (miR-34a, miR-34b, miR-34c) families show consistently negative correlations with the EMT score across all cancer types, implicating a role in suppressing EMT universally. On the other hand, miR-155 and miR-214 exhibited consistent positive correlations with the generic EMT score across all cancers, suggesting a role in promoting EMT universally. However, we also noted that miRNA previously reported to promote or suppress EMT (Hao *et al*, 2014; Zhang & Ma, 2012) were not in perfect concordance with our results (Fig S2). For example, miR-143, previously reported to suppress metastasis and stem cell characteristics (Hao *et al*, 2014; Zhang & Ma, 2012), had a positive correlation with generic EMT score, suggestive of a role in promoting EMT (Fig S2). This discrepancy could stem from the platform-specific, cross-hybridization problem related to the short, closely related nucleotide sequences between miRNA family members that makes it technically challenging to measure miRNA expression (Mestdagh *et al*, 2014).

This assessment is preliminary, as only small cohorts were analysed. Further investigations are required whenever more datasets including both miRNA and mRNA gene expression profiles become available.

Effect of reducing the number of genes in the generic EMT signature

If a biomarker or gene-based scoring system is to be quick, cheap and cost-effective in the clinical setting, the system needs to have as few biomarkers or genes as possible (CMTP, 2013). Thus, we sought to identify the utility and reliability of a smaller, generic EMT signature. To do this, we

applied an increasingly stringent degree of z-transformed weight thresholds and subsequently computed the EMT score of these new generic EMT signatures. The EMT score from these new signature was then correlated with the reference EMT score (Table E1A) across 17 cancer types (Table E4A) using the Spearman Correlation Coefficient test (Fig S5). The correlation decreased sharply to less than 0.85 when the z-score threshold was raised to above 5.0. At a threshold of 5.0, the number of genes in the generic EMT signature was 62 Epi and 24 Mes genes, which was much lower than the original 145 Epi and 170 Mes genes in the initial reference signature (corresponding to 57% and 86% reductions, respectively). When segregating a tumour to Epi, intermediate or Mes, the EMT scoring using these 86 genes was able to estimate EMT in the different cancer types with an overall concordance of 75.08% with reference to the full EMT score. We then repeated the same analysis for cell lines across 20 types of cancer (Tables E4B, E4C). Using the same threshold of 5.0, the number of genes significantly reduced to 87 Epi and 18 Mes genes, with an overall correlation of 0.88. Using this reduced signature for cell lines, we repeated the validation analysis presented in Fig S3 and Table E3. This reduced set accurately segregated Epi and Mes samples within the various datasets (Table E3). These results indicate that concordance between the full (tumour: 315 genes; cell line: 218 genes) and reduced (tumour: 86 genes; cell lines: 105 genes) EMT signatures. Notably, molecular diagnostics technologies are now capable of simultaneously assaying up to 800 genes (nCounter® from NanoString Technologies, Inc., Seattle, WA), which could be a potential platform for the proposed generic EMT scoring method using either the full or the reduced EMT signature.

Generic EMT signature and stemness

EMT is often associated with the acquisition of stemness (Frisch *et al*, 2013; Huang *et al*, 2012; Tam & Weinberg, 2013; Thiery *et al*, 2009), and therefore, we assessed if scoring with the generic EMT signature could also universally quantitate the stemness phenotype in different cancer types. Using Spearman Correlation Coefficient test, we computed the correlation of the generic EMT score and ssGSEA (Verhaak *et al*, 2013) enrichment score of 21 published stem cell-related signatures (*Beier Glioma stem cell*, *Ben-Porath embryonic stem cell v1*, *Ben-Porath embryonic stem cell v2*, *Ben-Porath NANOG targets*, *Ben-Porath NOS targets*, *Ben-Porath OCT4 targets*, *Ben-Porath SOX2*

targets, Bhattacharya embryonic stem cell, BIOCARTA stem pathway, Boquest stem cell cultured vs fresh, Boquest stem cell, Conrad stem cell, Gal leukemic stem cell, Gentles leukemic stem cell, Hoebeke lymphoid stem cell, Jaatinen hematopoietic stem cell, Lee neural crest stem cell, Oswald hematopoietic stem cell in collagen gel, Pece mammary stem cell, Wong embryonic stem cell core, Yamashita liver cancer stem cell) found in the Molecular Signature Database (Msigdb v4.0) (Subramanian *et al*, 2005), as well as published stem cell markers (Medema, 2013) (Fig S11). The generic EMT score does not correlate universally with stemness in all cancer types. Positive correlations between generic EMT score and stem cell signatures, such as *Boquest stem cell* and *Pece mammary stem cell*, or gene expression of stem cell markers, such as *CD44* and *CXCR4*, were observed in the majority of the cancer types. However, this result is limited by the fact that these stem cell signatures were derived from different cell types, with no consideration given to the different types of stem cells. For example, in breast cancer, there exist at least two types of breast cancer stem cells (Liu *et al*, 2014). Thus, although we observed a correlation of generic EMT score with stemness in cases, the correlation is not universal in all cancers or in all types of stem cell.

Supplementary Figure Legends

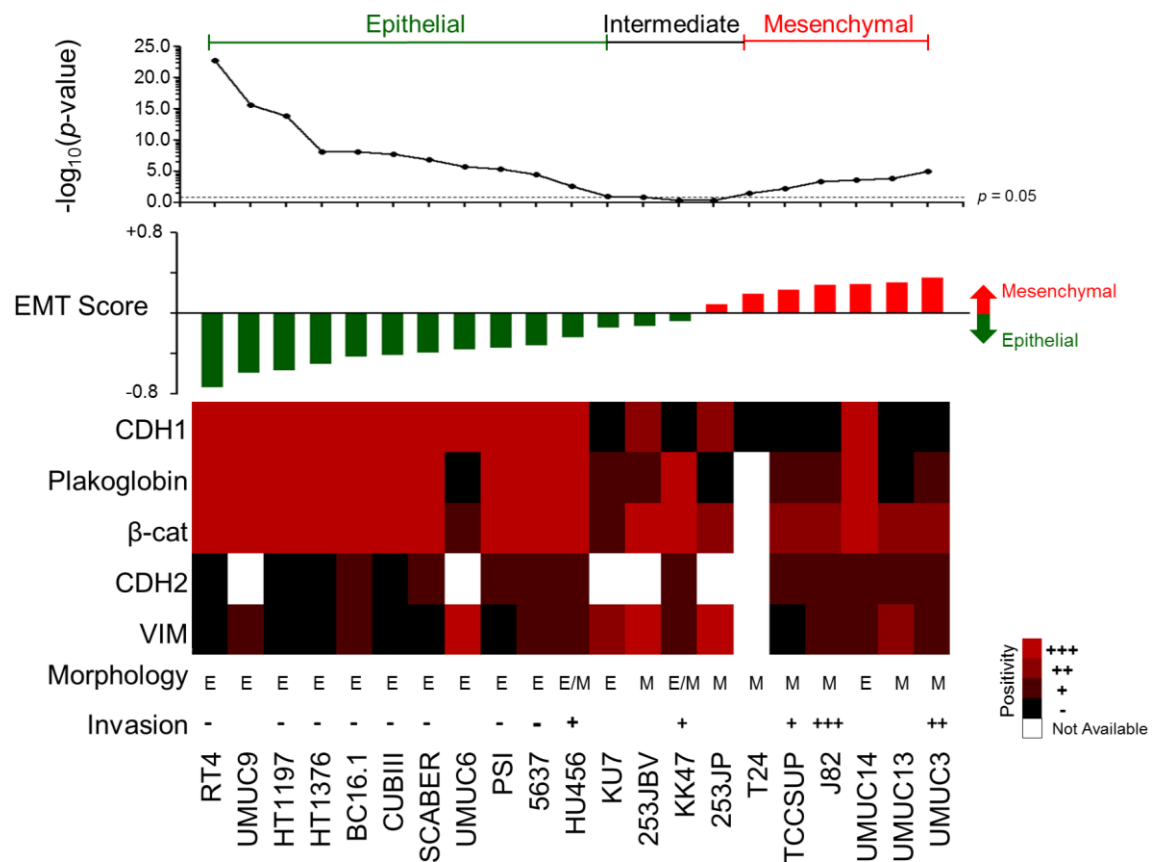


Figure S1: Validation of bladder-specific EMT signature.

Immunofluorescence staining heatmap of Epi (CDH1, Plakoglobin) and Mes (VIM, CDH2) markers, as well as non-nuclear β -catenin (black = low, red = high, white = no data). Bladder cancer cell lines ($n=21$) are aligned from the most Epi to most Mes based on the EMT score computed from gene expression data BLA-40 (Lee *et al*, 2007) and CCLE (Barretina *et al*, 2012), as shown by the bar chart. The dot plot is the $-\log_{10} p$ -value of two-sample Kolmogorov-Smirnov test. An arbitrary threshold of $p < 0.05$ was used to define Epi, intermediate, and Mes cell lines. Reported morphology (E=Epi, M=Mes) and 16-h invasion assay measuring number of cells that had traversed the membrane grid/mm (-, 0-4; +, 5-20; ++, 21-50; +++, >51 cells) were given. The staining, morphology and invasion data were taken or inferred from the literature (Baumgart *et al*, 2007; Black *et al*, 2008; Chen *et al*, 2009).

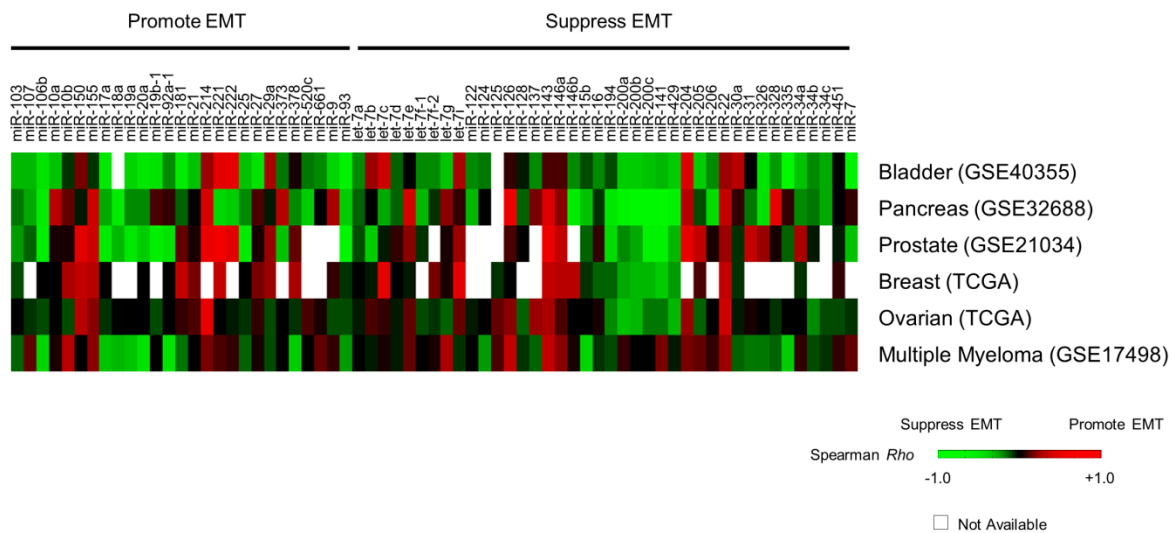


Figure S2: Correlation of generic EMT signature with miRNA implicated in EMT.

Heatmap of Spearman correlation coefficient Rho correlating generic EMT scores with miRNA expressions reported to regulate EMT.

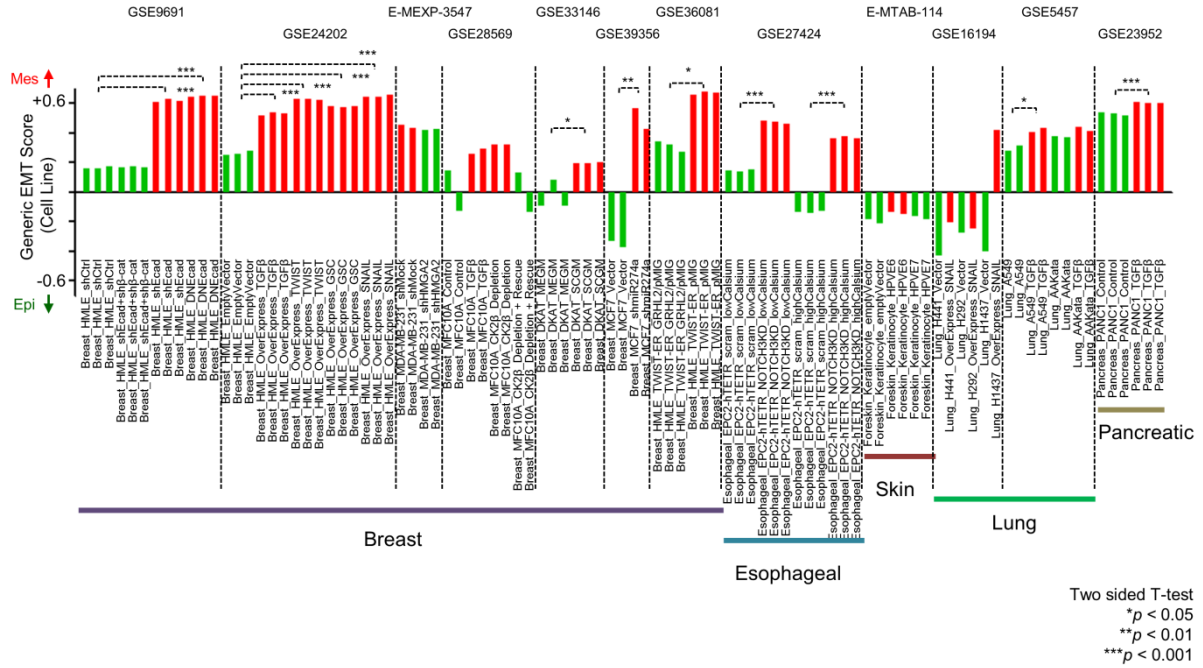


Figure S3: Validation of generic epithelial-mesenchymal transition (EMT) signature.

Bar chart of EMT scoring: in breast, GSE9691 (Onder *et al*, 2008), GSE24202 (Taube *et al*, 2010), E-MEXP-3547, GSE28569 (Deshiere *et al*, 2013), GSE33146 (D'Amato *et al*, 2012), GSE39356 (Cai *et al*, 2013), GSE36081 (Cieply *et al*, 2012); in oesophageal, GSE27424 (Ohashi *et al*, 2011); in skin, E-MTAB-114 (Hellner *et al*, 2009); in lung, GSE16194 (Yanagawa *et al*, 2009), GSE5457 (Malizia *et al*, 2009); and in pancreas, GSE23952 (Maupin *et al*, 2010). Dataset is separated by vertical dashed lines, with the dataset ID at the top of the chart. Cell line names and functional interventions are given at the bottom of chart. Coloured bars at the bottom-most region of the figure indicate the different cancer types. * indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$; computed by Mann Whitney U test. Colour code: green, more epithelial-like (Epi); red, more mesenchymal-like (Mes). EMT score and additional information are given in Tables E3 and E8.

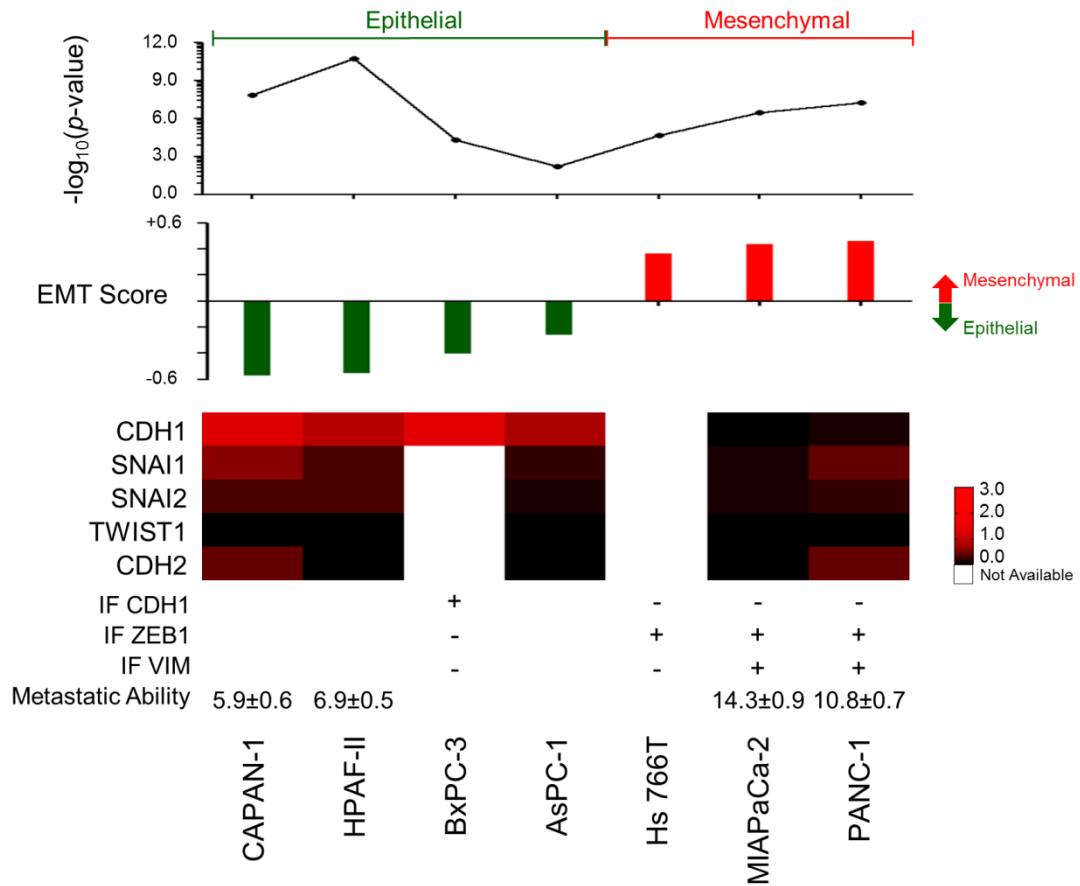


Figure S4: Validation of generic EMT signature on pancreatic cancer cell lines.

Normalized protein expression (relative to β -actin expression) estimated from western blot from the literature (Hotz *et al*, 2007; Zhu *et al*, 2012) for Epi (CDH1) and Mes (CDH2, SNAI1, SNAI2, VIM, and TWIST1) markers (black: no expression; red: high expression). Top panel shows the two-sample Kolmogorov-Smirnov test p -value for EMT scoring. Second panel is the EMT score computed using the generic EMT signature and data from SANGER COSMIC (Garnett *et al*, 2012) and CCLE (Barretina *et al*, 2012). The metastatic ability (in nude mouse) and immunofluorescence (IF) staining (-: no expression, +: positive) data was adapted from the literature (Arumugam *et al*, 2009).

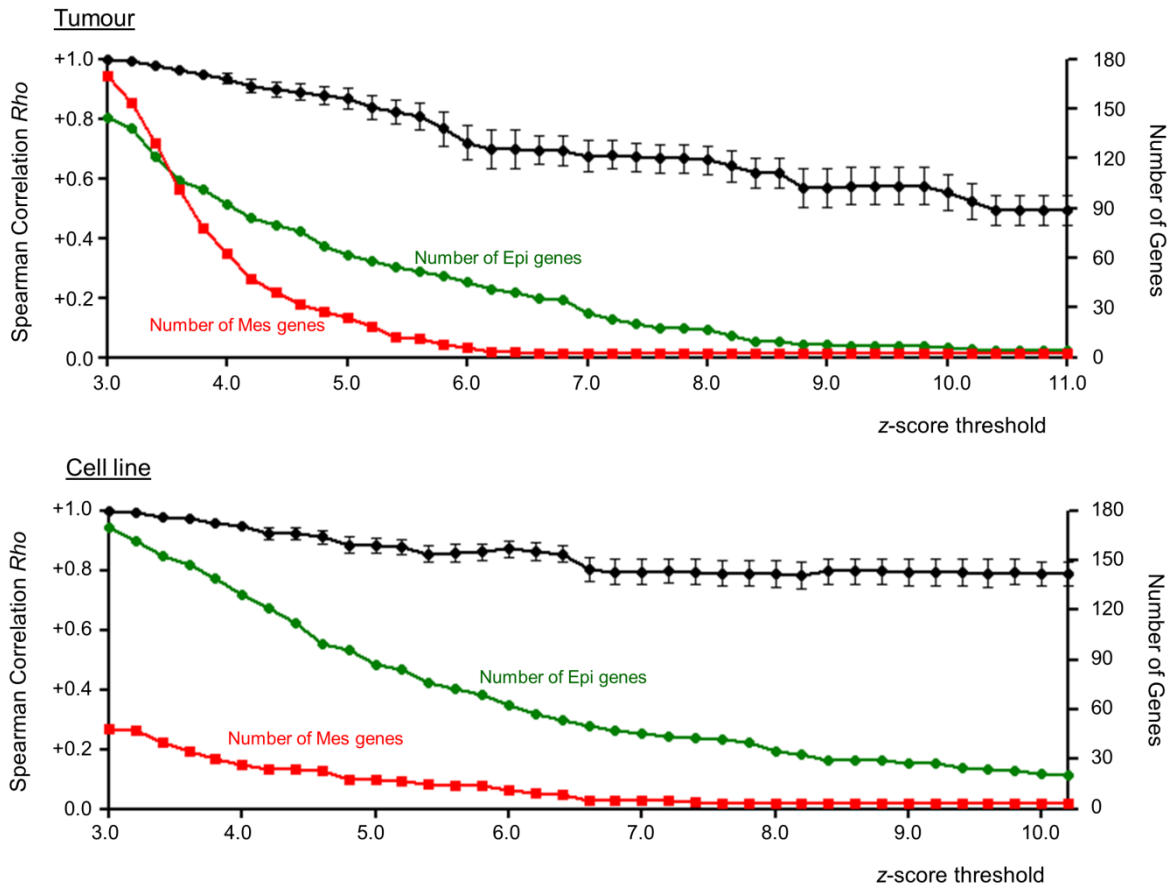


Figure S5: Effect of reducing genes in generic EMT signature.

Plot of Spearman correlation coefficient Rho (mean $\pm$ SEM from 17 cancer types in tumour and 20 cancer types in cell lines; black; left y-axis), as well as number of genes in generic EMT signature (Epi genes, green; Mes genes, red; right y-axis) against the z-transformed weight threshold (x-axis), computed based on Significance Analysis of Microarray (SAM) fold-change, false discovery rate, Receiver Operating Characteristics (ROC), and number of samples of a gene in each cancer-specific EMT signature.

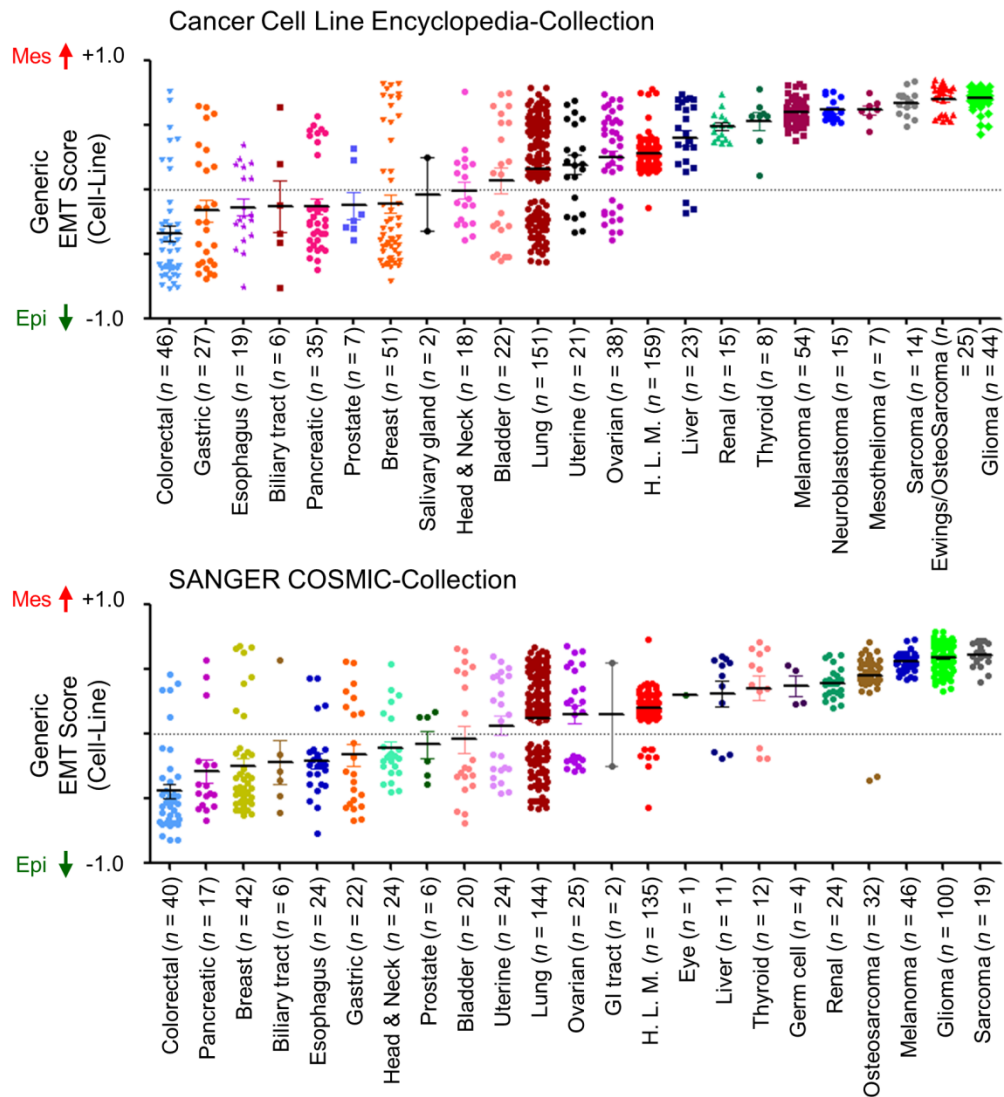


Figure S6: Epithelial-mesenchymal transition (EMT) spectrum in different cell lines.

Scatter plot of EMT scores for various cancer cell lines sourced from the Cancer Cell Line Encyclopedia (CCLE) (Barretina *et al*, 2012) (upper panel) and SANGER COSMIC (Garnett *et al*, 2012) (lower panel) cell line collections. Cell lines are grouped according to cancer/lineage and the groups are sorted by mean EMT score, as indicated by the horizontal centre line. The bars above and below this line indicate the $\pm$ square error of the mean. EMT score nearer to +1.0 is more mesenchymal-like (Mes) whereas EMT score nearer to -1.0 is more epithelial-like (Epi). Abbreviations: H. L. M., hematopoietic/lymphoid malignancy.

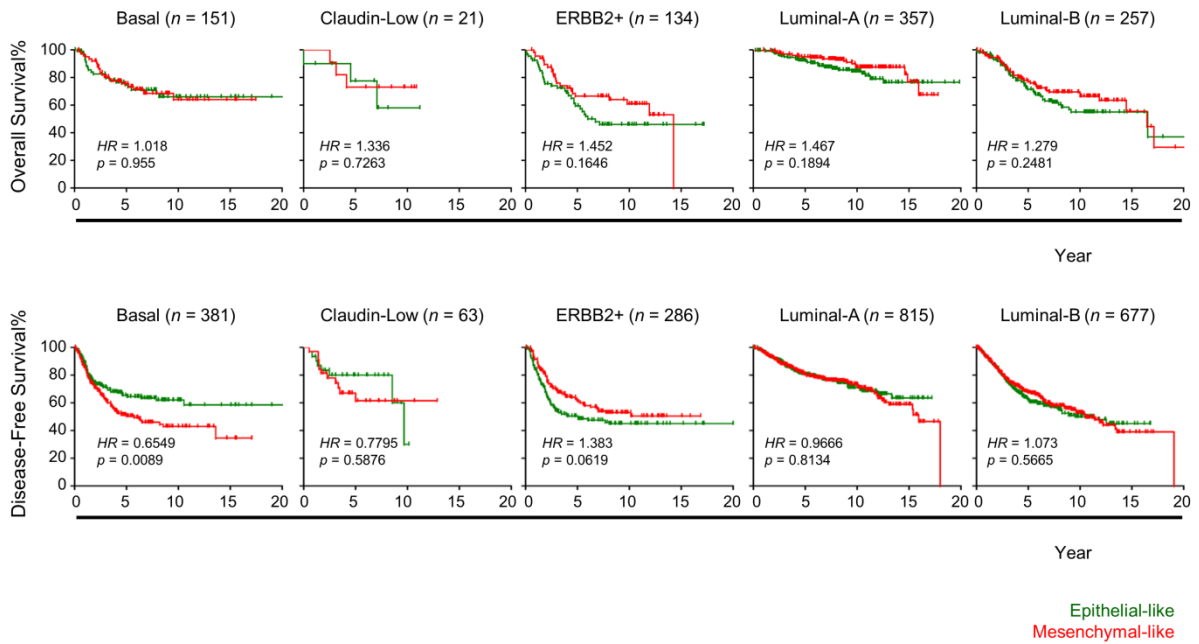


Figure S7. Correlation of Epithelial-Mesenchymal Transition and survival in breast cancer subtypes.

Kaplan–Meier analysis of overall survival (upper panel) and disease-free survival (lower panel) comparing mesenchymal-like (Mes; red) and epithelial-like (Epi; green) breast cancers in five molecular subtypes: Basal, Claudin-Low, ERBB2+, Luminal-A and Luminal-B. The molecular subtypes were predicted based on a subtype signature (Prat *et al*, 2010) and ssGSEA (Verhaak *et al*, 2013), as described in (Akalay *et al*, 2013). Median of EMT score was used to categorize breast cancer into Mes ($\geq$ median) or Epi ($<$ median). Hazard ratio (HR) and log-rank test p -values are shown. The number of samples (n) is given in parentheses.

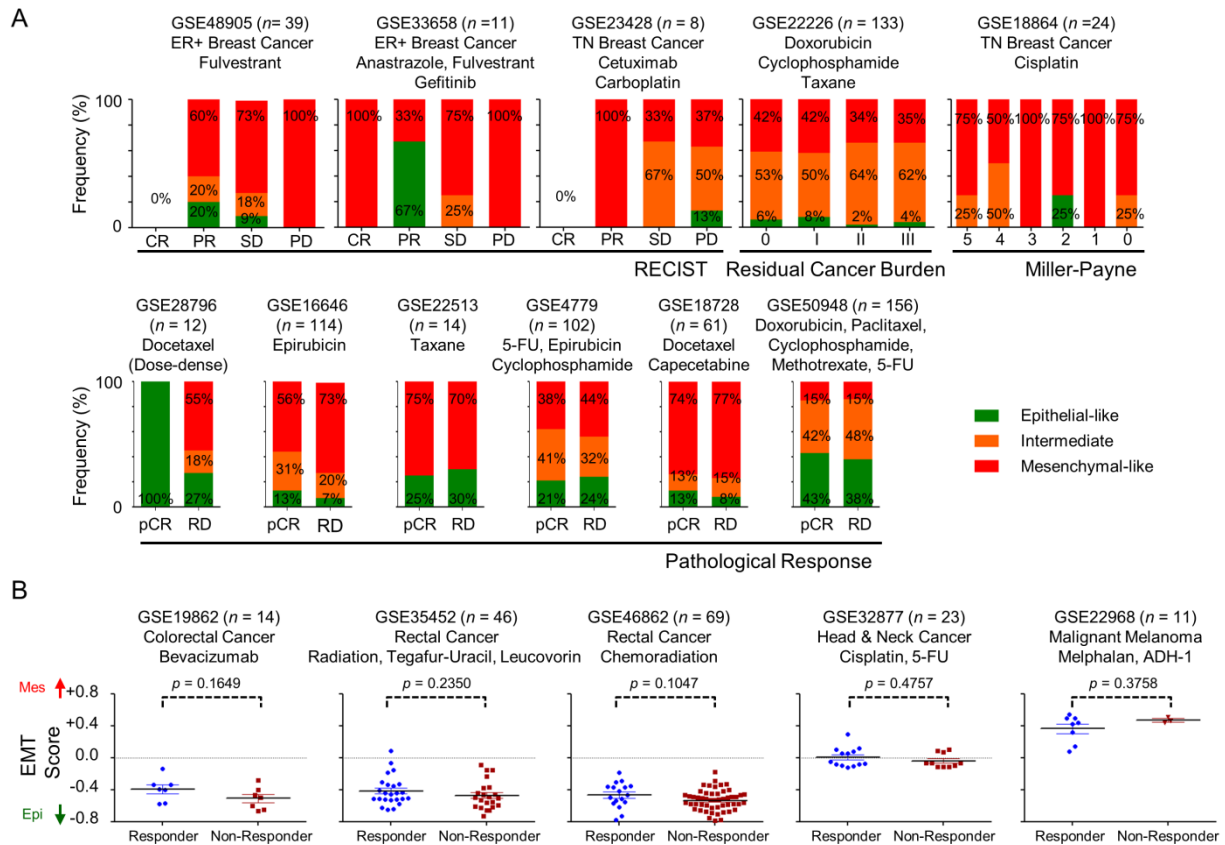


Figure S8. Generic epithelial-mesenchymal transition (EMT) and drug sensitivity.

A. Bar plots of 11 breast cancers cohorts (Bauer *et al*, 2010; Carey *et al*, 2012; Esserman *et al*, 2012; Evans *et al*, 2012; Farmer *et al*, 2009; Knudsen *et al*, 2014; Korde *et al*, 2010; Lehmann *et al*, 2011; Massarweh *et al*, 2011; Prat *et al*, 2014; Silver *et al*, 2010) and EMT status in clinical response to treatment. Percentage distributions of EMT status are given in each clinical response group. The bars are aligned from most sensitive to most resistant according to different assessment criteria (response evaluation criteria in solid tumour (RECIST), residual cancer burden (RCB) grading, Miller-Payne grading, and pathological response), as given in the original publications. The number of samples is given in parentheses, and the neoadjuvant treatment regimen is also indicated. Cohorts comprise a mix (ER+, HER2+ or ER-/PR-/HER2-) of breast cancer subtypes, unless specified. Abbreviation: CR, complete response; PR, partial response; SD, stable disease; PD, progressive disease; pCR, pathologic complete response; RD, residual disease; TN, triple negative; 5-FU, Fluorouracil. Green, epithelial-like (Epi); orange, intermediate; red, mesenchymal-like (Mes).

B. Dot plot of EMT score (mean $\pm$ SEM; y-axis) for responders (blue) and non-responders (maroon) in 3 recurrent colorectal cancer cohorts (Gim J, 2014), 1 head & neck cancer cohort (Tomkiewicz *et al*, 2012), and 1 malignant melanoma cohort (Beasley *et al*, 2011). Clinical response was evaluated based on RECIST or radiological examination, and the *p*-value was evaluated using Mann-Whitney *U*-test.

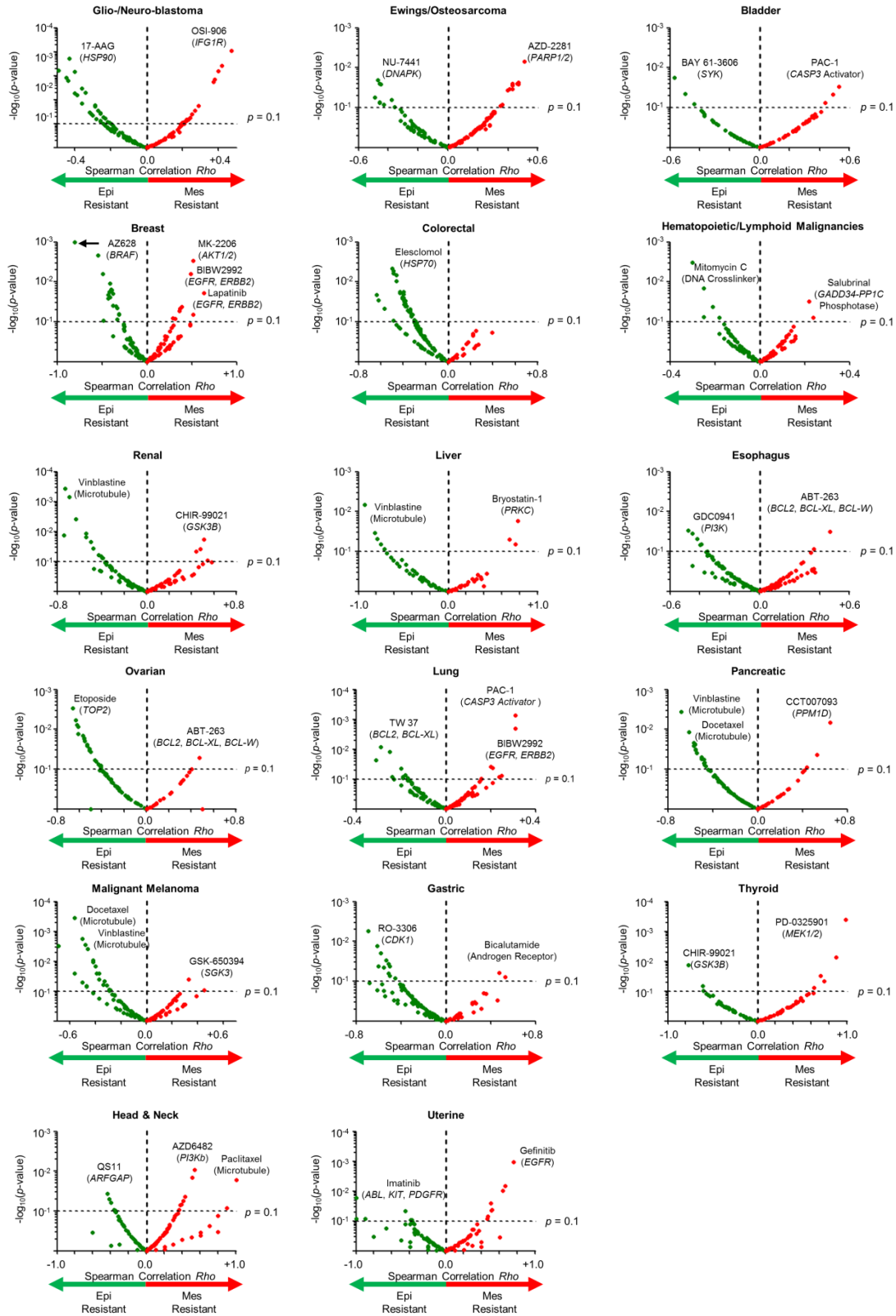


Figure S9: Epithelial-mesenchymal transition (EMT) and drug sensitivity.

Volcano plot of EMT correlation with drug sensitivity in 17 different cancers. $Rho \in [-1, +1]$ (x-axis) and $-\log_{10} p\text{-value}$ (y-axis) were computed by Spearman correlation coefficient test. A dashed line corresponding to $p=0.1$ is plotted. Red indicates higher drug resistance in mesenchymal-like

(Mes) tumours ($Rho \in [0, +1]$), whereas green indicates higher drug resistance in epithelial-like (Epi) tumours ($Rho \in [-1, 0]$). Selected compounds are labelled in different plots.

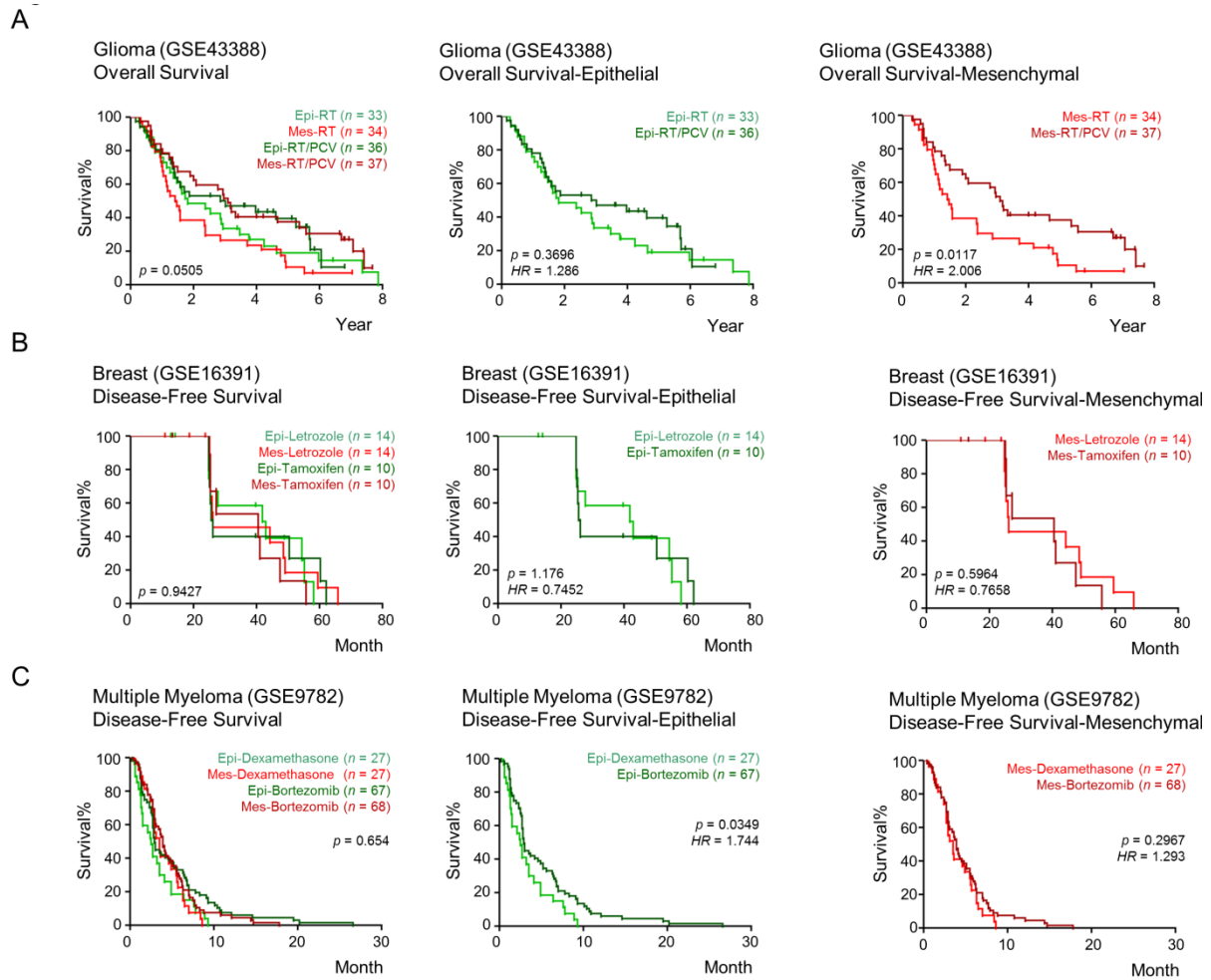


Figure S10. Differential treatment response in Epi and Mes tumours.

Kaplan-Meier analyses of overall survival or disease-free survival in patients with (A) Epi and Mes glioma (GSE43388) (Erdem-Eraslan *et al*, 2013) who underwent radiotherapy alone (RT) or radiotherapy and procarbazine, vincristine (RT/PCV), (B) ER+ breast cancer (GSE16391) (Desmedt *et al*, 2009), who underwent letrozole or tamoxifen, and (C) multiple myeloma (GSE9782) (Mulligan *et al*, 2007), who underwent dexamethasone or bortezomib. Left panels are the combined survival plots; middle panels are the survival plots of Epi tumours, and right panels are the survival plots of Mes tumours. Colour code: Green, epithelial-like (Epi); Red, mesenchymal-like (Mes). Light or dark colour indicates different treatment regimens. Epi and Mes are defined based on median EMT score, and p -value was evaluated using the log-rank test.

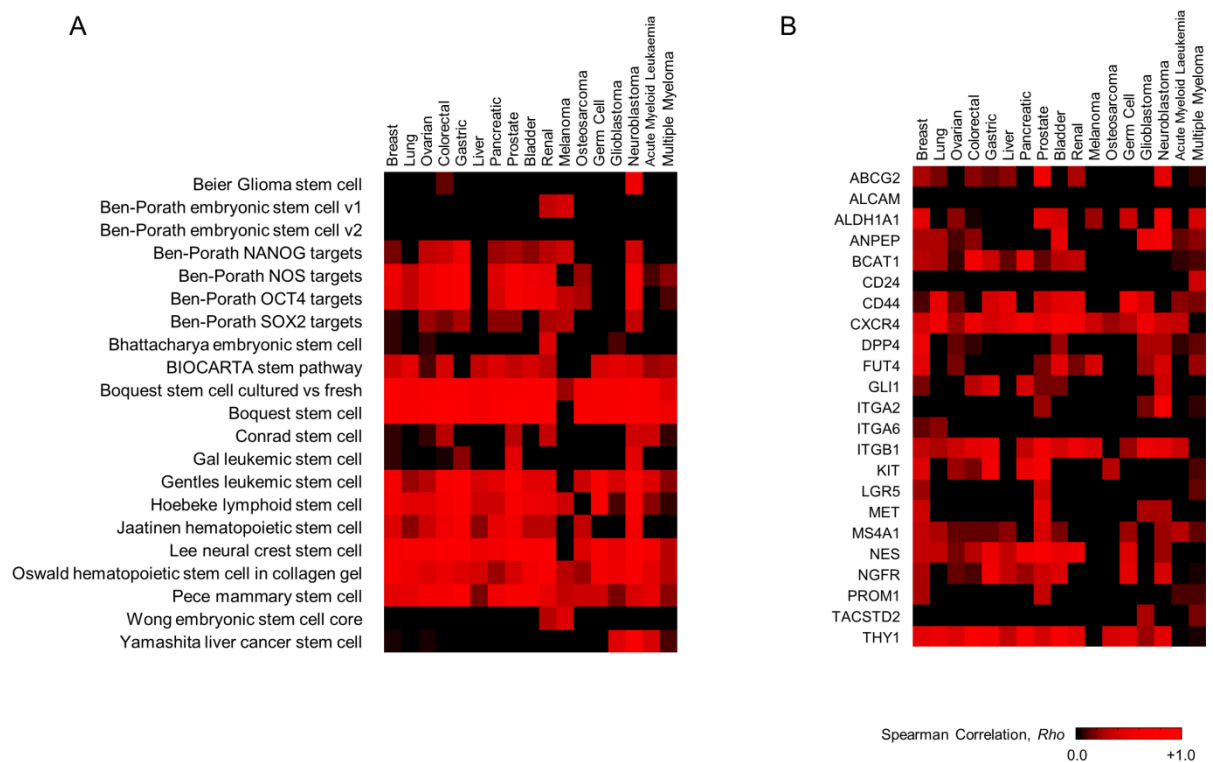


Figure S11: Correlation of generic EMT signature and stemness.

Heatmap of Spearman correlation coefficient Rho (red: positive correlation, black: no correlation) showing correlations of the generic EMT signature and the enrichment score of stemness-related gene sets from (A) Molecular Signature Database v4.0 (Subramanian *et al*, 2005) and (B) published stem cell markers.

Supplementary Materials and Methods

Quality control of Affymetrix microarray expression data

The quality of the Affymetrix chip (Affymetrix, Santa Clara, CA) was confirmed using the Bioconductor AffyQCReport package (Gautier *et al*, 2004) and the following criteria: average perfect-match intensity, kernel density plot, GAPDH 3':5' ratio, β -actin 3':5' ratio, and centre of intensity for positive and negative controls. All chips passed at least one of the criteria, and hence, none of the samples was discarded.

Clinical tissue samples of Japanese Foundation for Cancer Research

Clinical tissues were obtained from patients with breast cancer who had never received chemotherapy prior to sample collection and who underwent surgical resections of malignant lesions at the Cancer Institute Hospital, Japanese Foundation for Cancer Research (JFCR; Tokyo, Japan) between 2001 and 2007. This study was approved by the institutional review board at JFCR and informed consent was obtained from each patient. Each tissue sample was immediately embedded in a plastic cryomold containing Tissue-Tek OCT compound (Sakura Finetek, Tokyo, Japan), frozen with liquid nitrogen and kept in a -145°C freezer. Frozen tissue specimens were cut (10- μ m-thick) in series using a cryostat (CM3050S, Leica Microsystems, Wetzlar, Germany) and placed onto plain, uncoated glass slides and stored at -20°C. Subsequently, sections were stained with haematoxylin and eosin and covered by micro-cover glasses. The prepared slides were pathologically examined to distinguish clusters of mammary tumour cells from normal and stromal cells.

RNA isolation and microarray analysis

Frozen OCT-embedded tissues were washed with 70% ethanol and water, and stained with Mayer's haematoxylin for 30 sec. Stained sections were water-washed and then dried for the subsequent dissection process. Cancer cells were isolated by laser capture micro-dissection (LCM, Leica Microsystems). Total RNA was extracted using RNeasy Micro Kit (Qiagen, Hilden, Germany)

according to the manufacturer's procedure. The total RNA concentration for each sample was measured using a NanoDrop spectrophotometer (NanoDrop Technologies, Wilmington, DE). Any possible degradation of the total RNA was validated using a BioAnalyzer 2100 (Agilent Technologies, Santa Clara, CA). Total RNA with the desired concentration and quality was amplified and biotinylated with GeneChip Two-Cycle Target Labeling and Control Reagents (Affymetrix). Concentration and size of cRNA were validated with the BioAnalyzer. Biotinylated cRNA was fragmented following the Two-Cycle assay protocol. The fragmented cRNA was validated in terms of fragment size, again using the BioAnalyzer, and applied to GeneChip Human Genome U133 Plus 2.0 (Affymetrix) and incubated at 45°C for 16 h. After hybridization, the microarray was washed and stained with a streptavidin phycoerythrin (SAPE) conjugate using a fully automated system (GeneChip Fluidic Station 450; Affymetrix), according to the manufacturer's procedure. The hybridized probe arrays were scanned using the GeneChip Scanner 3000 (Affymetrix). The microarray data were deposited at Gene Expression Omnibus (GEO) with the accession number GSE54002.

Breast Cancer Subtype Prediction

For each breast cancer sample, subtype signature (Prat *et al*, 2010) scores for Basal, Claudin-Low, Luminal, ERBB2+, and Normal-like subtypes were computed using single-sample Gene Set Enrichment Analysis (ssGSEA) (Verhaak *et al*, 2013). The subtype of each breast cancer sample was inferred based on the maximum subtype signature score. For Luminal breast cancers, the PAM50 signature (Parker *et al*, 2009) was applied to further classify them into Luminal-A and Luminal-B subtypes.

Predictive modelling and validation by BinReg

Expression data analysis based on a binary regression model using the BinReg v2.0 (renamed to Profiler, <http://dig.genome.duke.edu/software.html>) has been described previously (Gatza *et al*, 2010; Tan *et al*, 2013). Briefly, BinReg uses a binary probit regression model and Bayesian statistics to produce fitted regression coefficients on a set of genes that are most correlated with the binary

response/phenotype of interest (e.g., epithelial vs. mesenchymal). The regression coefficients of these genes indicate the discriminating power of the genes and are weights for the overall meta-gene profile. The overall meta-gene profile is used for comparison and predicts the status of the EMT phenotype of the new sample or dataset. To predict the status of the phenotype on a dataset, a Bayesian probit regression model was fit to assign the probability that the sample exhibited evidence of a phenotype, based on the concordance of its gene expression values with the signature (Gatza *et al*, 2010).

Protein expression quantification from western blot

Western blot images were downloaded from the journal website (Hotz *et al*, 2007) and processed using ImageJ 1.46r software (NIH, Bethesda, MD). A rectangular selection tool was used to mark the regions of interest, and quantitation was performed using the area under the curve (AUC) of the intensity histogram. Regions of interest with no unimodal peaks in the intensity histogram were considered as ‘no expression’. The AUC was normalized to that of β -actin and the normalized AUC was taken as the protein expression.

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